Bleeding and Thrombotic Complications

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40.1 Introduction

Bleeding and thrombotic complications are an important cause of morbidity and mortality in patients undergoing HCT. The major thrombotic complications include venous thromboembolism (VTE) including catheter-related thrombosis (CRT), sinusoidal obstruction syndrome (SOS), and transplant-associated thrombotic microangiopathy (TA-TMA), while bleeding commonly involves the gastrointestinal or respiratory tracts and is most common in thrombocytopenic patients or those with GVHD. HCT is associated with multiple risk factors for both thrombosis and bleeding including the underlying malignancy, thrombocytopenia, high-dose myeloablative chemotherapy (MAC) and immune-modulatory drugs, GVHD, infections, indwelling vascular catheters, and prolonged immobilization (Chiu and Lazo-Langner 2023; Gerber et al. 2008; Chaturvedi et al. 2016; Nadir and Brenner 2007). In addition, HCT is also associated with alterations in the coagulation system with activation of endothelium-dependent coagulation factors, increase in von Willebrand factor (vWF) and platelet adhesion, increased thrombin generation, decreased antithrombin levels, and decreased levels of anticoagulant proteins such as protein C (Vannucchi et al. 1994). Collectively, major patient-, disease-, and therapy-related factors contribute to hemostatic complications in HCT patients. Thrombotic and bleeding complications in HCT are discussed separately in the following section.

Over the past few years, chimeric antigen receptor T-cell (CAR-T) therapy has revolutionized the field of malignant hematology, particularly for relapsed and refractory B-cell lymphomas, acute lymphoblastic leukemia, and multiple myeloma (MM). Along with these advances, bleeding and thrombosis have emerged as important toxicities associated with CAR-T therapy and are attributed to slow hematopoietic reconstitution, hemostatic defects, and a tendency to thrombosis due to thromboinflammatory influences that are potentiated in the setting of cytokine release syndrome or disseminated intravascular coagulation. We also review the emerging evidence regarding recognition and management of CAR-T-associated bleeding and thrombotic complications in the following section.

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40.2 Thrombotic Complications in HCT

40.2.1 Epidemiology and Risk Factors

Thromboembolic complications in HCT recipients include venous thromboembolism (VTE), catheter-related thrombosis (CRT), sinusoidal obstruction syndrome, and TA-TMA. VTE is the most common of these complications, and retrospective studies have reported VTE incidence ranging from 4.6% over 180 days in patients undergoing HCT (Gerber et al. 2008). The rate of VTE is higher with allo-HCT than auto-HCT and in the presence of GVHD with 1-year VTE rates of 4.8%, 6.8%, and 8.1% reported with auto-HCT, allo-HCT without GVHD, and allo-HCT with GVHD, respectively (Pihusch et al. 2002). A retrospective series of 447 patients undergoing bone marrow transplantation reported a 5.7% incidence of VTE in the first 100 days following transplant despite being on heparin prophylaxis (100 U/kg iv daily) for hepatic SOS (Pihusch et al. 2002). Finally, Gonsalves et al. reported a 1-year symptomatic VTE incidence of 3.7% in patients undergoing HCT in an *ambulatory* care setting (Gonsalves et al. 2008).

VTE occurs most frequently following engraftment in patients undergoing allogenic HCT and those with a history of previous VTE or GVHD (Labrador et al. 2013; Gerber et al. 2008). An increased risk of VTE can be observed in both acute and chronic GVHD and is the result of systemic inflammation and endothelial injury (Chiu and Lazo-Langner 2023). The majority of VTE episodes in these studies were catheter-related thromboses. Cortelezzi et al. have previously reported a 12% incidence of catheter-related thromboembolic complications in a cohort of 416 patients with hematologic malignancies (Cortelezzi et al. 2005). Twenty-one percent of these patients were HCT recipients, and 81.2% had platelet counts less than 50×10^9/L. There was a nonstatistically significant trend toward lower rates of thrombotic complications with thrombocytopenia. In a report that included 1514 patients undergoing inpatient HCT, 70 patients (4.6%) had 75 symptomatic VTE events, of which 55 (73.3%) were catheter-associated. In the same cohort, clinically significant bleeding occurred in 230 patients (15.2%; 95% CI, 13.4–17.1%); 55 patients (3.6%; 95% CI, 2.7–4.7%) had fatal bleeding (Gerber et al. 2008). Overall, in individuals undergoing HCT, the risk of significant bleeding appears to be greater than the risk of serious complications from thrombosis, which greatly influences the weighing of risk versus benefits of thromboprophylaxis, particularly in the setting of thrombocytopenia.

40.2.2 VTE Prophylaxis

40.2.2.1 Randomized Studies

Randomized studies have not evaluated empiric prophylactic anticoagulation in HCT recipients; however, studies in patients with cancer provide the next best evidence that can be extrapolated. The PROTECHT (nadroparin versus placebo) and SAVE-ONCO (semuloparin versus placebo) trials showed a significant reduction in the relative risk of venous thromboembolism with prophylactic anticoagulation in patients with cancer; however, the absolute risk reduction is small, and no survival benefit has been demonstrated. The American Society of Clinical Oncology (ASCO) guidelines advise against the use of routine prophylactic anticoagulation in ambulatory patients with cancer (Key et al. 2020). We do not generally recommend prophylactic anticoagulation in thrombocytopenic HCT recipients with the exception of those with multiple myeloma (MM) receiving thalidomide or lenalidomide or hospitalized patients at higher risk of thrombosis (Table 40.1).

40.2.2.2 Multiple Myeloma

Patients with multiple myeloma have a high baseline risk of thrombosis of 5–10% which increases several-fold in patients being treated with the immunomodulators (IMiDs) thalidomide and lenalidomide with dexamethasone or chemotherapy. Consolidation therapy with the thalidomide or lenalidomide after HCT has been shown to improve complete remission rates and

Table 40.1 Recommendations for prophylaxis and treatment of VTE in HCT recipients[a]

VTE prophylaxis	VTE treatment
Indications for prophylaxis – Patients with MM receiving IMiDs – During hospitalization or postoperatively, as long as platelet count is >50 × 10^9/L. – Prophylaxis is not recommended in outpatients with indwelling vascular catheters **Prophylaxis strategy** – Aspirin in low-risk patients with MM receiving IMiDs – LMWH (prophylactic dose of 40 mg SC daily) for patients with MM on IMiDs and >1 risk factor for VTE – Prophylactic doses of UFH or LMWH in hospitalized patients	**General principles** – Start therapeutic doses of LMWH or IV UFH in patients who have platelet count >50 × 10^9/L and no active bleeding. UFH is preferred in case of renal impairment (GFR <30 mL/min) or high bleeding risk – Continue LMWH or transition to warfarin (if LMWH is contraindicated) for maintenance therapy – DOACs are not currently recommended in patients undergoing HCT **Duration of anticoagulation** – *General*: 3–6 months or as long as malignancy or use of IMiDs persists, whichever is longer – *Catheter-related thrombosis*: 3 months or as long as catheter is in place **Inferior vena cava filter** Only use to patients in whom anticoagulation is contraindicated, or those who develop pulmonary embolism on anticoagulation. Remove as soon as anticoagulation can be started

DOACs, direct oral anticoagulants; *IMiDs*, immunomodulatory drugs; *LMWH*, low-molecular-weight heparin; *MM*, multiple myeloma; *UFH*, unfractionated heparin
[a]Adapted from Chaturvedi et al. (2016)

prolong event-free survival and is thus rapidly becoming the standard of care (McCarthy et al. 2012; Barlogie et al. 2006). In patients receiving thalidomide consolidation after auto-HCT for MM, the rate of VTE was 24% and 6% in the induction and consolidation periods, respectively, despite thromboprophylaxis with low-molecular-weight heparin (LMWH) (Barlogie et al. 2006). McCarthy et al. reported no episodes of VTE in patients receiving consolidation therapy with lenalidomide; however, these patients also received prophylactic anticoagulation (McCarthy et al. 2012). Based on studies showing the benefit of thromboprophylaxis in patients with newly diagnosed multiple myeloma receiving lenalidomide- or thalidomide-based treatment (Palumbo et al. 2011) and the ASCO recommendation for thromboprophylaxis in this population (Lyman et al. 2015; Key et al. 2020), we recommend either aspirin or LMWH for lower risk patients and LMWH for higher-risk patients receiving thalidomide or lenalidomide.

40.2.2.3 Hospitalized Patients

Though there is a clear benefit of pharmacologic thromboprophylaxis in medically ill hospitalized patients (Samama et al. 1999), randomized trials have not evaluated thromboprophylaxis in HCT patients. The potential benefit from VTE prophylaxis is proportional to VTE risk, and therefore this is particularly important in patients with reduced mobility and with a history of VTE (if not on long-term anticoagulation) due to an even higher risk of thrombosis. Our practice is to start prophylactic anticoagulation for hospitalized patients in the posttransplant period once the platelet count is >50 × 10^9/L, and there is no active bleeding. For very high-risk patients, anticoagulation can be considered if the platelet count is >30 × 10^9/L; however, this must be balanced with the risk of bleeding.

Risk models developed recently have sought to predict VTE risk after allogeneic transplantation to inform which patients may benefit from VTE prophylaxis after platelet engraftment. Martens et al. developed the HIGH-2-LOW score which is calculated from seven predictors (history of CRT, inpatient at day 30, GVHD grade 3 or 4, history of PE or lower extremity DVT, lymphoma diagnosis, BMI ≥ 35, and WBC count ≥11 × 10^9/L) evaluated at day 30 after allogenic transplant that each contributes one point to stratify patients into low (0 points), intermediate (1 point), or high risk (≥2 points) for VTE between day 30 and 100 after allogenic transplant (Martens et al. 2021). Patients with a high, intermediate, and low risk had a 10.3%, 3.6%, and 1.5% risk of

incident VTE, respectively, between day 30 and day 100.

The HiGHS2 score is another risk model that evaluates VTE risk at 2 years after transplantation and includes the history of stroke (3 points), chronic GVHD (1 point), hypertension (2 points), male sex (2 points), peripheral blood stem cell source (3 points) as predictors that classify patients into high (≥5 points) and low VTE (<5 points) risk, with a 9.3% and 2.4% incident risk of 10-year VTE, respectively (Gangaraju et al. 2021).

40.2.2.4 Prophylaxis of Catheter-Related Thrombosis

HCT patients, especially those undergoing ambulatory HCT, frequently have indwelling vascular catheters with the potential of catheter-related thrombosis (CRT). Despite multiple randomized and observational studies, thromboprophylaxis for the prevention of catheter-related thrombosis in patients with cancer remains controversial. The largest study of thromboprophylaxis in central venous catheters randomized 1590 cancer patients undergoing chemotherapy to adjusted-dose warfarin (international normalized ratio, 1.5–2.0), fixed-dose warfarin (1 mg/day), and no prophylaxis (Young et al. 2009). Symptomatic CRT was less frequent in the patients given adjusted-dose warfarin than in those who received no prophylaxis (2.7% versus 5.9%, $P = 0.019$); however, both adjusted-dose and fixed-dose warfarin were significantly associated with increased risk of major bleeding (Young et al. 2009). Recent meta-analyses of randomized trials concluded that prophylactic warfarin and LMWH do not significantly reduce symptomatic CRT in patients with cancer (Akl et al. 2007). Based on the available evidence, we do not routinely recommend prophylactic anticoagulation to prevent catheter-related thrombosis.

40.2.3 VTE Diagnosis and Treatment

Venous duplex ultrasonography should be performed in patients presenting with extremity swelling, redness or tenderness, or pulmonary angiography in patients with chest pain, dyspnea, or unexplained tachycardia. A clinical assessment of bleeding risk is necessary in patients who are diagnosed with VTE. Patients with no increased risk based on bleeding history and platelet count >50 × 10^9/L should be started on therapeutic anticoagulation with either LMWH or unfractionated heparin (UFH). The use of LMWH is restricted to patients with glomerular filtration rate > 30 mL/minute, while UFH is used in patients with impaired renal function (glomerular filtration rate < 30 mL/min) or those with high bleeding risk. Following initiation of anticoagulation with LMWH or UFH, patients may be continued on LMWH or transitioned to an oral anticoagulant such as warfarin or possibly a direct oral anticoagulant. The direct oral anticoagulants (DOACs) have not been evaluated specifically in HCT recipients; however, given their widespread use in clinical practice, and data supporting their use in patients with malignancy, they may be considered on a case-by-case basis in patients who have platelet counts stably over >50 × 10^9/L, no recent bleeding, and adequate renal function. The optimal duration of anticoagulation for VTE in HCT patients has not been evaluated in prospective studies. The recommendation for patients with cancer-related VTE is anticoagulation for 3–6 months, with ongoing therapy if the malignancy persists (Key et al. 2020; Kearon et al. 2012). We follow an analogous strategy in HCT patients with the caveat that extended anticoagulation is often not feasible in patients with relapsed disease and a high likelihood of disease-related or treatment-related thrombocytopenia (Table 40.1).

The use of inferior vena cava (IVC) filters should be restricted to patients with acute deep vein thrombosis and a contraindication to anticoagulation and possibly patients who develop pulmonary embolism while on therapeutic anticoagulation (Kearon et al. 2012). IVC filters should not be used for primary prophylaxis of pulmonary embolism. In patients with large, symptomatic thrombosis and severe thrombocytopenia, we sometimes follow a strategy of platelet transfusions to reach a threshold of 50 × 10^9/L to allow safer anticoagulation with heparin. In the

setting of thrombocytopenia with platelets 25–50 × 10^9/L, some centers have used reduced-dose LMWH for VTE treatment and may be a viable strategy after weighing the patient-specific benefits of anticoagulation with the risks of bleeding (Ibrahim et al. 2005; Lam et al. 2021; Mantha et al. 2017).

40.2.4 Treatment of Catheter-Related Thrombosis

The rate of PE and mortality from CRT is low, and the objectives of CRT treatment are to reduce symptoms, prevent extension into more central veins, preserve access, and prevent chronic venous stenosis. There is no evidence that removal of the catheter improves outcomes. Therefore, it is reasonable not to remove the catheter unless it is nonfunctional, is no longer needed, or may be infected. Thrombus reduction by catheter-directed thrombolysis is relatively safe and effective and may be tried in an attempt to preserve the catheter. Anticoagulation is required in patients with acute CRT regardless of whether the catheter is removed (Kearon et al. 2012; Lyman et al. 2015). We prefer LMWH, though vitamin K antagonists (VKA) may be used if LMWH is contraindicated. In a prospective study of 78 patients with CRT treated with full-dose dalteparin bridged to warfarin, there were no new thrombotic events at 3 months and 57% of catheters were still functional (Kovacs et al. 2007). The optimum duration of anticoagulation has not been evaluated in prospective studies. Current ACCP guidelines recommend anticoagulation for 3 months or until the catheter is removed, whichever is longer (Kearon et al. 2012). Several clinicians prefer to continue anticoagulation for 1–2 weeks after the catheter is removed.

40.2.5 Sinusoidal Obstruction Syndrome (SOS)

SOS is a life-threatening complication that presents usually within the first 45 days after HCT with elevated serum bilirubin levels, painful hepatomegaly, and fluid retention (Carreras 2015). Endothelial injury of the hepatic sinusoids in SOS initiates hepatocyte injury and liver failure. SOS can occur in as high as 8–13% of HCT recipients, and mortality is in excess of 80% (Carreras 2015). Myeloablative conditioning, preexisting liver disease, younger age, and poor performance status are associated with an increased risk of SOS (McDonald et al. 1993). Ursodeoxycholic acid is recommended as prophylaxis for SOS in patients undergoing allo-HCT. Anticoagulation with low-dose heparin has also been studied and is sometimes prescribed to patients undergoing auto-HCT. Defibrotide, a pro-fibrinolytic agent, is a new agent approved for the treatment of severe SOS in both children and adults and is associated with higher rates of survival than historical controls (20–30% at day 100) (Richardson et al. 2016). Defibrotide prophylaxis has been shown to have some efficacy in preventing SOS in high-risk children, but whether this benefit translates for adults is not known.

40.2.6 Transplant-Associated TMA

TA-TMA is a heterogenous, frequently fatal disorder that occurs within 100 days after HCT and is caused by treatment- and disease-related endothelial damage, coagulation activation, and microvascular thrombosis (Nadir and Brenner 2007). It is characterized by thrombocytopenia, microangiopathic anemia with schistocytes on the blood smear, and varying organ impairment such as renal failure and neurological symptoms. The diagnosis can be challenging, since the clinical symptoms overlap with other common complications including GVHD and infections (Rosenthal 2016). Risk factors for developing TA-TMA include exposure to calcineurin inhibitors, high-dose chemotherapy, GVHD, infections, advanced age, female sex, and non-MAC (Elsallabi et al. 2016). Elevated levels of vWF and inflammatory mediators, such as IL-1, TNF-alpha, and thrombomodulin, and neutrophil extracellular traps have been implicated as causing the endothelial damage in TA-TMA. Treatment of TA-TMA is mostly supportive; however,

recent data show that some patients with severe TA-TMA harbor complement gene mutations, and uncontrolled complement activation has been demonstrated in TA-TMA, which is a potential therapeutic target. The complement inhibitor eculizumab has been successfully used in some cases of TA-TMA (Rosenthal 2016).

40.3 Bleeding Complications

Bleeding in HCT recipients is closely associated with prolonged and severe thrombocytopenia. In retrospective studies, the rate of bleeding in HCT recipients ranges from 15.2% to 27.1%, and life-threatening or fatal bleeding occurred in 1.1% to 3.6% of patients (Gerber et al. 2008; Pihusch et al. 2002; Labrador et al. 2013). Gerber et al. reported that the initiation of therapeutic anticoagulation during days 1–180 after HCT was the strongest predictor of bleeding [OR 3.1 (95% CI 1.8–5.5)] (Gerber et al. 2008). Furthermore, GVHD [OR 2.4 (95% CI 1.1–3.3)] increased the risk of bleeding, while auto-HCT (versus allo-HCT) was protective [OR 0.46 (95% CI 0.33–0.64)]. Bleeding can take any form, including gastrointestinal hemorrhage in patients with GVHD of the gut, hemorrhagic cystitis in patients with genitourinary involvement by GVHD, viral reactivation, and alkylating agent therapy, or spontaneously. Diffuse alveolar hemorrhage (DAH) is a devastating bleeding complication that occurs in 2%–14% of HCT recipients and presents with progressive hypoxia, pulmonary infiltrates, and bloody alveolar lavage (Nadir and Brenner 2007). DAH is more common in thrombocytopenic patients and those with acute GVHD, and the effects of inflammatory cytokines on the alveolar lining have been implicated. There are no evidence-based prophylactic and therapeutic strategies, and reported mortality is around 80% (range 64 to 100%) (Afessa et al. 2002). Platelet transfusions, systemic corticosteroids, antifibrinolytics, and recombinant factor VIIa have all been used with inconsistent results. It is general practice to administer prophylactic platelet transfusions for platelet counts less than 10×10^9/L in patients undergoing myeloablative chemotherapy or HCT, though the superiority of prophylactic over therapeutic platelet transfusions is supported by low- to moderate-grade evidence. Given the competing risks of bleeding and thrombosis, identifying patients at high risk for these outcomes can optimize strategies for prophylaxis. The timing of hemostatic complications is an important consideration, since bleeding events are more likely to occur early in the post-transplant course when patients are profoundly thrombocytopenic, while thrombotic events occur more frequently after hematopoietic recovery (Gerber et al. 2008; Labrador et al. 2013).

40.4 Thrombosis and Bleeding in CAR-T

Thrombotic and bleeding events are also common complications seen in patients after chimeric antigen receptor T-cell (CAR-T) therapy. In a retrospective report of 148 patients receiving CD19 CAR T-cell therapy for large B-cell lymphoma, the incidence of VTE was 11% between day 0 and day 100 of CAR T-cell infusion (Hashmi et al. 2020). Half of all new thrombotic events were deep vein thromboses and half of which were catheter-related. Twenty-five percent were pulmonary emboli and the remaining quarter were categorized as other (mesenteric, cerebral, and renal). Risk factors associated with VTE were bulky disease, use of bridging therapy, worse performance status, severe cytokine release syndrome, and immune effector cell-associated neurotoxicity syndrome (ICANS). Most of these patients with VTE were treated with anticoagulation, either heparin products or direct oral anticoagulants, which was later held if thrombocytopenia developed. There were no major bleeding events or deaths from VTE or bleeding.

Another retrospective study by Parks et al. reported a 9% incidence of VTE within 60 days of CAR-T cell infusion in 91 patients with relapsed/refractory non-Hodgkin lymphoma or multiple myeloma (Parks et al. 2021). Of these patients, a majority were started on anticoagulation with direct oral anticoagulants or LMWH, one of whom had to stop therapy due to thrombo-

cytopenia. None experienced bleeding events or recurrent thromboses.

In a cohort of 127 patients, Johnsrud et al. found that 6.3% of patients developed VTE and 9.4% developed bleeding complications within the first 3 months after CAR T-cell infusion (Johnsrud et al. 2021). Bleeding events included gross hematuria, soft tissue bleeding, gastrointestinal hemorrhage, hemoptysis, and subdural hematoma. Older age, elevated pre-lymphodepleting chemotherapy lactate dehydrogenase, and lower baseline platelet count were associated with bleeding. Patients with bleeding also had lower platelet and fibrinogen nadirs compared to patients without bleeding events. High-grade (≥3) ICANS was associated with both bleeding and thrombosis, and CRS was not associated with either.

There are no large, prospective studies that evaluate prophylaxis and management of hemostatic complications in patients undergoing CAR T-cell therapy. Preliminary reports indicate that thrombotic and bleeding events are common, that anticoagulation for the treatment of VTE may be safe when platelets are above 50×10^9/L, and that hemostatic complications may be correlated with other CAR-T adverse events such as cytokine release syndrome, neurotoxicity, thrombocytopenia, and hypofibrinogenemia.

Key Points

- Hemostatic complications, including both thrombosis and bleeding, are common in HCT recipients and contribute to morbidity and mortality.
- Indwelling vascular catheters, GVHD-associated inflammation, and certain medications are important risk factors for VTE, while prolonged severe thrombocytopenia and GVHD predispose to bleeding.
- Pharmacologic thromboprophylaxis is recommended for patients with MM receiving IMiDs and hospitalized patients with platelet count >50×10^9/L, but not for routine prophylaxis of CRT.
- LMWH (or UFH) is the treatment of choice for VTE in HCT recipients.
- Ursodiol and defibrotide are recommended for the prevention and treatment of SOS, respectively. Defibrotide may also have a role in the prophylaxis of high-risk patients.

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